

Monitoring water availability is crucial in India as it can help to address climatic and environmental challenges. Information on water bodies can help to understand the temporal changes taking place in the environment and the climate, and to get a glimpse of the potential drivers of such changes.

The development of the lake management and monitoring system (LMMS)

Gowtham Gollapalli, founder of Kaiinos Geospatial Technologies, the company that has developed a lake management and monitoring system for the state of Telangana, says, "Mapping and conservation of water bodies is an important task for citizen groups and the government. So, we started this journey with a citizen group (SOUL or Save Our Urban Lakes) in Hyderabad to map all the water bodies that are present there. During those interactions we met government officials, and they were interested to map the water bodies of the entire state."

So Kaiinos Geospatial Technologies, along with IIIT-Hyderabad and the government of Telangana, began work on an open source lake management and monitoring system (LMMS) that would map and monitor the changes in the water bodies in Hyderabad over a period of time, using satellite data and location based apps on the Web and mobile.

Mission Kakatiya is an initiative by the Telangana state government to restore minor irrigation sources of the region. As almost 85 per cent of the cultivated area in Telangana is rain fed, lakes are very crucial for sustaining agriculture in the region. The government is restoring all the lakes in the state for sustainable overall development in a phased manner.

As per a technical document released by the Telangana government, "The project aims to use the rich satellite data sets and derived data sets provided by the Indian Space Research Organization (ISRO) for improving the efficiency in planning, monitoring and implementing Mission Kakatiya. Rich spatial data acts as a backbone for the proposed lake information system."

A three-pronged approach is used to extract the spatial data, by using the multispectral satellite data provided by NASA and ISRO, and analysing the natural water sinks extracted from a combination of satellite data, contour data and digital elevation models.

The information extracted from digital/vector formats of the toposheets of Survey of India is also utilised for creating highly accurate and location-specific information about the lakes. The officials of the irrigation department are provided with a user interface to harvest the information so that they can plan, monitor and implement lake regeneration with ease and efficiency.

What is the need for mapping water bodies?

Gollapalli says that mapping of water bodies is needed to understand the alterations that have taken place over a period of time due to demographic or climatic changes. Experts can then get to know the impact of these changes on agriculture and drinking water, and find out the areas that are under water stress (urban or rural).

